

The Quest to Build a Quantum Computer

The road to large-scale quantum computing is long and hard, with incremental advances paving the way. But the destination is in sight.

Quantum computing stands at the cutting edge of modern technology, promising transformative advances in everything from cryptography to drug discovery. At the heart of this revolution is the hardware—the physical machinery that enables quantum computers to operate. The global race to build quantum computers is not just about speed or size; it is a contest to master the most delicate and challenging aspects of quantum physics, making it one of the most exciting frontiers in science and engineering.

From silicon to superconductors: The evolution of quantum hardware

Silicon has long been the backbone of classical computers, powering everything from laptops to supercomputers. When physicists and engineers first ventured into the world of quantum computing, it was natural to start with what they knew best—silicon. Early quantum devices attempted to adapt silicon transistors, hoping to harness quantum effects like superposition and entanglement within a familiar framework. While these initial attempts provided important insights and proved that quantum behaviour could be observed in solid-state systems, the limitations of silicon soon became apparent. Issues such as decoherence—where quantum states lose their delicate properties—posed



significant challenges, making it difficult to maintain stable qubits for practical computations.

Recognising the obstacles posed by silicon, researchers began exploring other materials that might better support quantum phenomena. This led to experiments with semiconductors such as gallium arsenide, and even exotic systems like trapped ions and photons. Each approach brought its own advantages and drawbacks. For instance, ion-trap and photonic quantum computers offered excellent

coherence times and control, but scaling these systems for large-scale computations proved tricky. Meanwhile, semiconductor-based quantum dots allowed precise control over electrons, pointing towards new possibilities for qubit design.

As quantum computing research matured, superconducting materials emerged as a breakthrough technology. Superconductors are remarkable because, at extremely low temperatures, they allow electrical current to flow without any resistance. This property is ideal for creating quantum circuits, where maintaining quantum states with minimal interference is essential. Superconducting qubits, often crafted from materials like aluminium and niobium, are now the foundation of many leading quantum computers.

Superconducting qubits are typically implemented using devices such as Josephson junctions, which exploit quantum tunnelling effects to create robust, controllable qubits. These systems offer fast operation speeds and relatively straightforward integration with existing electronics, making them highly attractive for commercial and research purposes. Companies and institutions around the world—Google, IBM, and others—have built